Table S1. Primers and enzymes used for experimental verification of candidate imprinted genes and controls. All primers are listed in 5-3 direction. SNPs are highlighted in red in restriction enzyme recognition sites and in CDS sequence context. For reference the SNP is listed within its sequence context of plus/minus 6 bp.

primer name	primer sequence	fragment size	Restriction Enzyme	recognition site	digestion products	Ecotype	SNP +/- 6nt
CSN2_fwd	TCGAGTATTCTGATGAGGACCAAG	318	DdeI	CTGAG	114+204	Sn	TGCTGAAGTTGTT
CSN2_rev	TAGCAGATCCAGAGACGAAATCC					Pp	TGCTGAGGTTGTT
FIS2_fwd	CAAGGGCCTGCAACTTCATTTG	501	BanI	GGTGCC	93+408	Sn	TCCCAAAGTGCCT
FIS2_rev	ACAACATCATCATCATTCACACGAG					Pp	ACCTAAGGTGCCT
HAIKU1-fwd	AGACACAGCCTCAGGTTTATAAC	318	FauI	CCCGC	119+199	Sn	ACAACCTGCCAAA
HAIKU1-rev	CAGCTGTATTAGACCAAAACTGG					Pp	ACAACCCGCCAAA
jg11174-fwd	ACACCAAAACCATCGGATCAAC	455	HaeIII	GGCC	173+282	Sn	GAAATTAGCCCGG
jg11174-rev	AGCTCCAATGGTAGACTCTACTATG					Pp	GAAATTGGCCCGG
jg14499_fwd	GATCCTATCGACAAACTATTCAAGG	354	BanI	GGTACC	116+238	Sn	CGGTACGTCATTA
jg14499_rev	ACGATACGAACTTGCCGATAG					Pp	CGGTACCTCATTA
PEG5_fwd	AAGTGTCTTTGGTGGTACTTTGG	244	BsaI	GGTCTCA	90+154	Sn	TTGGTCACAAAAC
PEG5_rev	GCGAATCATATCTGCTTGTTGGAC					Pp	TTGGTCTCAAAAC
PKR2_fwd	TGGTTATGTACACTGGGACTTCG	453	Hin1I	GACGTC	100+353	Sn	CCTCCTGACGTCG
PKR2_rev	GCATCTTATCCTTGAGCACGTC					Pp	CCTCCTCACGTCG
AGL62_fwd	ACATGCAACTTAGTGAGACTCG	434				Sn	TCAACTCGATGGA
AGL62_rev	GTTTCCAAACGGTAATGCGGG					Pp	TCAACTGGATGGA
CSN6B-fwd	ATGGCGCCTTCTTCGTCGAG	457				Sn	CCTCCCTCCGCTT
CSN6B-rev	GATCCTTCTGTGCATGATTGATAGC					Pp	CCTCCCGCCGCTT
GAPDH-fwd	TGCTTACAGTCATGATGAACCC	541				Sn	AGGTTGCTAGACG
GAPDH-rev	ACCCCATTCATTATCATACCAAGC					Pp	AGGTTGTTAGACG
jg14501_fwd	GAAAACGGCAGAAGCAGCAAG	306				Sn	TCTTCCTTACTAC
jg14501_rev	TAGGAAAGGAGTAGTTGTTGCGG					Pp	TCTTCCATACTAC
jg16612-fwd	GGAGGCATTTCTTCTTCTCAGTG	666				Sn	ATCTCACGGTGTA
jg16612-rev	ACCATTCATGTTTCACAGCATCC					Pp	ATCTCATGGTGTA
PEG4_fwd	GAACACCGAGCTATTCCTCTCTG	505				Sn	AAGAATCAAATTC
PEG4_rev2	GAACCGCGAGTAATTAGTCTTGTG					Pp	AAGAATTAGAATC
PEG6_fwd	CAAAGACACGGAAGGACTAGGTC	220				Sn	TTTGTCTTAGATC
PEG6_rev	AGCATCATAGTAAAGAGCTGCCT					Pp	TTTGTCCTAGATC

Table S2. Qualitative comparison of 3 ASE detection methods. Rows 1-3 emphasize that K-IRP omits the two align-to-reference steps required by the other methods. Instead, K-IRP tests whether read pairs have one or more k-mers specific to one parent and none specific to the other. **Rows 4-6** compare overall qualities. All three methods rely on an annotated reference for binning reads by gene, but K-IRP uses a k-mer based mapper and is therefore alignment-free.

Method	Description
1. SNP-counting methods	1. Align both parent RNAseq datasets to the reference 2. Create catalogs of parent-specific (homozygous) SNPs 3. Align cross reads to the reference 4. Assign cross reads to parent by counting SNPs per read
2. IRP	1. Align parent parent RNAseq datasets to the reference 2. Create (haploid) parent-specific consensuses by polishing methods 3. Align cross reads both parent-specific consensuses 4. Assign cross read to parent by comparing quality of its parental alignments
3. K-IRP	1. Aligning parent reads to the reference is not required 2. Create catalogs of parent-specific k-mers (removing k-mers in the intersection and low-frequency k-mers likely induced by sequencing error) 3. Aligning cross reads to the reference is not required 4. Assign read-to-parent based on counting its parent-specific k-mers
4. Gene bins	All three methods use an annotated reference to bin cross reads per gene. SNP-counting and IRP use a sequence alignment tool. K-IRP uses an alignment-free, k-mer based mapping tool. All three methods only report on genes that show insufficient reads per experiment. IRP and KIRP add pseudocounts of one to avoid division by zero in fold change calculations.
5. Gene filter	IRP and KIRP run simulations to avoid false positives. They mix parental RNAseq data in equal proportions and process the combination as if it were cross RNAseq. Any genes showing allelism by this test are removed from the study.
6. Two way	All three methods use reciprocal crosses (swapping the maternal and paternal roles of both ecotypes) to avoid false positives. They only report genes that are consistently a MEG or a PEG under both conditions.

Table S3. Case-wise comparison of 3 ASE detection methods. The 7 rows show all possible combinations of diploid parental genotypes (blue, red) at a single nucleotide position. These include two homozygous parents (rows 1,2), one heterozygous parent (rows 3,4), and two heterozygous parents (rows 5,6,7). The specific nucleotides shown in column 1 illustrate the general cases. Each method may correctly use (green), misinterpret (yellow), or ignore (white) the position. All the potential false positive cases (yellow) would fail to pass the reciprocal cross test. Florez-Rueda et al. 2016 and Roth et al. 2018 introduced a SNP-counting variant that specifically addresses the case in row 3, one parent is heterozygous and shares a nucleotide with the other parent, which is homozygous. This table interprets one position, but all three methods combine information from all the variant positions traversed by a read pair.

Parent Genotypes	SNP counting	IRP	KilRP
1 A A A A	No contribution to ASE detection.	No contribution to ASE detection.	No contribution to ASE detection.
2 A A C C	Position contributes to ASE detection.	Position contributes to ASE detection.	Position contributes to ASE detection.
3 A A A C	Florez-Rueda extrapolates from the number of **C** reads.	Potential false positive if consensuses differ.	Potential false positive: assigns **C** reads but not **A** reads.
4 A A C G	No contribution to ASE detection.	Position contributes to ASE detection (with half counts for parent 2).	Position contributes to ASE detection.
5 A C A C	No contribution to ASE detection.	Potential false positive if consensuses differ.	No contribution to ASE detection.
6 A C C G	No contribution to ASE detection.	Potential false positive if consensuses differ.	Position contributes to ASE detection (with half counts).
7 A C G T	No contribution to ASE detection.	Position contributes to ASE detection (with half counts).	Position contributes to ASE detection.